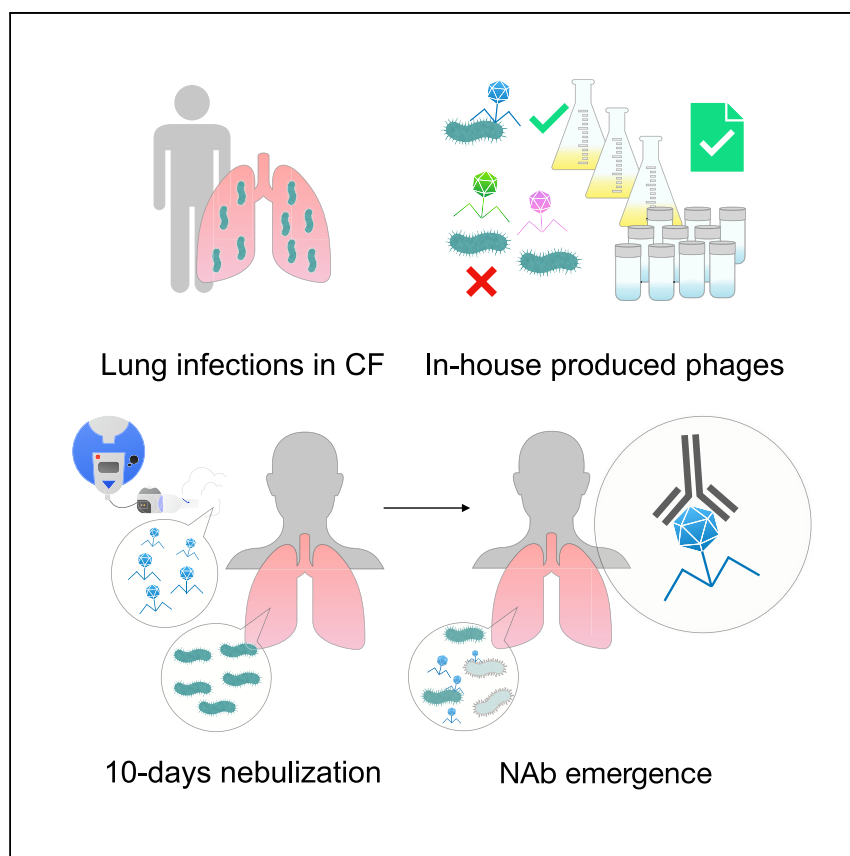


Case Report

Neutralizing antibodies after nebulized phage therapy in cystic fibrosis patients



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Highlights

First report of anti-phage-neutralizing antibodies after nebulized therapies

Long-term follow-up is crucial; antibodies are detected up to 42 days after treatment

The generation of antibodies does not prevent a favorable clinical outcome

Phages can remain in the body after treatment is ceased

Bernabéu-Gimeno et al. describe three nebulized phage therapy treatments in two patients with cystic fibrosis. The authors report for the first time the emergence of anti-phage antibodies after nebulized phage therapy, demonstrating the immunogenicity of phages regardless of the route of administration.



Translation to Patients

Bernabéu-Gimeno et al., Med 5, 1096–1111
September 13, 2024 © 2024 The Author(s).
Published by Elsevier Inc.
<https://doi.org/10.1016/j.medj.2024.05.017>



Case Report

Neutralizing antibodies after nebulized phage therapy in cystic fibrosis patients

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SUMMARY

Background: Cystic fibrosis (CF) patients are prone to recurrent multi-drug-resistant (MDR) bacterial lung infections. Under this scenario, phage therapy has been proposed as a promising tool. However, the limited number of reported cases hampers the understanding of clinical outcomes. Anti-phage immune responses have often been overlooked and only described following invasive routes of administration.

Methods: Three monophage treatments against *Staphylococcus aureus* and/or *Pseudomonas aeruginosa* lung infections were conducted in cystic fibrosis patients. In-house phage preparations were nebulized over 10 days with standard-of-care antibiotics. Clinical indicators, bacterial counts, phage and antibiotic susceptibility, phage detection, and immune responses were monitored.

Findings: Bacterial load was reduced by 3–6 log in two of the treatments. No adverse events were described. Phages remained in sputum up to 33 days after completion of the treatment. In all cases, phage-neutralizing antibodies were detected in serum from 10 to 42 days post treatment, with this being the first report of anti-phage antibodies after nebulized therapy.

Conclusions: Nebulized phage therapy reduced bacterial load, improving quality of life even without bacterial eradication. The emergence of antibodies emphasizes the importance of long-term monitoring to better understand clinical outcomes. These findings encourage the use of personalized monophage therapies in contrast to ready-to-use cocktails, which might induce undesirable antibody generation.

Funding: This study was supported by the Spanish Ministry of Science, Innovation and Universities; Generalitat Valenciana; and a crowdfunding in collaboration with the Spanish Cystic Fibrosis Foundation.

INTRODUCTION

The emergence and spread of multi-drug-resistant (MDR) bacteria are major health threats worldwide.¹ Due to the lack of effective antibiotic therapies, phages (viruses infecting bacteria) have been proposed as a therapeutic alternative.² Phages are the most abundant biological entities, showing an enormous diversity and high specificity. Their use in clinics has been tested since their discovery, although they are currently in the spotlight due to the rise of MDR bacterial infections. However, the lack of regulatory approval in most countries restricts phage therapy to compassionate use, making it difficult to determine the criteria for production, dosage, administration, and monitoring. In addition, comparison of approaches between centers and countries is challenging, making critical the need of standardized

CONTEXT AND SIGNIFICANCE

Multi-drug-resistant (MDR) bacteria are a global health threat, especially for cystic fibrosis (CF) patients with recurrent lung infections. Phages are a promising alternative, but the limited number of clinical trials and the lack of standardized monitoring protocols are an obstacle to their routine use in clinics. The authors describe phage therapy treatments in patients with CF, reporting for the first time the emergence of anti-phage-neutralizing antibodies after nebulization. Their results highlight the importance of assessing the immunogenicity of phage therapeutics regardless of route of administration and the use of personalized phage therapy versus ready-to-use cocktails. This work contributes to the understanding of phage therapy and remarks on the need to regulate and implement phage therapy through clinical trials.

monitoring protocols to further understand the effectiveness and outcome of these new treatments.³

Prior to administration, the first crucial step is to select the phage (or phages) efficiently targeting the bacteria to treat. Despite the significant biodiversity of phages, not all of them have the potential to be implemented in clinics, and they must be thoroughly characterized prior to their use.⁴ Determination of their life cycle is mandatory, since only strictly lytic phages (avoiding potential gene transfer) should be considered for therapy. In addition, bacterial defense mechanisms⁵ can reduce the therapeutic effect of phages since it has been shown that phage resistance readily emerges. However, it can be prevented or delayed by different approaches. A common method is the combination of phages in a cocktail, allowing them to further reduce the bacterial burden and to increase the host range.⁶ Indeed, this strategy has been widely employed as ready-to-use products for prevention or therapeutics.⁷ Another approach is combining phages with antibiotics, since it has been shown that phage-resistant mechanisms could re-sensitize bacteria to antibiotics, known as phage-antibiotic synergy (PAS effect).⁸

A major concern with phage therapy is the eventual immune response against phages.⁹ Despite recent publications in the field, the determinants of phage immunogenicity remain unclear. In some phage treatments, an immune response has been detected, while other works have not been able to demonstrate this, possibly due to the assumption that phages do not generate a generalized immune response.^{10,11} It is worth noting, however, that most cases differ in phage combination, dosage, treatment duration, delivery strategy, and immunological status of the patient, making it difficult to understand the impact of the immune response on treatment outcome.^{10,11} Indeed, given the natural interaction between humans and phages,^{12,13} the relevance of immune response evaluation has often been overlooked, and the assessment of anti-phage antibody production following phage therapy has not been systematically examined. Despite this, it has been suggested that the route of phage administration could have an impact on immune response generation, with potent responses more associated with invasive delivery strategies.^{10,11}

The consequences of the absence of available treatments to combat MDR bacteria are more relevant in patients with underlying specific conditions such as immunosuppression or cystic fibrosis (CF).¹⁴ CF is a chronic multisystemic genetic disease that primarily affects the respiratory, digestive, and reproductive systems.¹⁵ The reduced fluidity of secretions and consequent impaired mucociliary function makes the respiratory tract of CF patients subject to recurrent and chronic bacterial infections from an early age.^{16,17} Consequently, CF treatments often involve long-term antibiotic therapies, including a combination of antibiotics to combat bacterial exacerbations.¹⁴ *Staphylococcus aureus* and *Pseudomonas aeruginosa* are among the most prevalent pathogens associated with pulmonary infections in CF patients,¹⁸ both presenting a high level of antibiotic resistance.^{19,20} Indeed, *S. aureus* is the most common isolated bacteria in the CF respiratory tract at early ages.¹⁸ Infections by *S. aureus* can favor further colonization by other bacteria, specifically *P. aeruginosa*, which is the most prevalent pathogen in CF adults.²¹

Two pediatric case reports have been previously described, both colonized by *S. aureus* and *P. aeruginosa*. They were treated with a commercial phage cocktail, Pyophage (PYO), targeting simultaneously the two bacteria.^{22,23} In one of the cases, the cocktail was enriched with Sb1 *S. aureus* phage in a second round of therapy.²² Both showed a significant reduction in bacterial counts in sputum and improved

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<https://doi.org/10.1016/j.medj.2024.05.017>

Table 1. Case report summary

Patient	Bacteria	Key clinical data	Phage	NAb detection	Microbiology outcome	Clinical outcome
LFV	<i>S. aureus</i>	predominant infection; treated with oral antibiotics, mainly linezolid	Mallokai	day 10	short-term: >5 log CFU reduction, followed by increase of <i>P. aeruginosa</i> counts; long-term: return to pre-treatment values	stable; no adverse effects
	<i>P. aeruginosa</i>	secondary infection; treated with nebulized colistin or tobramycin	LPS-5	day 17	no change	stable; no adverse effects
HVR	<i>P. aeruginosa</i>	treated with nebulized colistin + aztreonam or levofloxacin; IV antibiotic therapy every 1–2 months	10	day 42	short-term: >6 log CFU reduction; long-term: >2 log CFU reduction	overall improvement; discontinued IV antibiotic therapy; no adverse effects

For each patient, clinical status, bacterial infection treated, and main results from follow-up after the monophage treatments are described. NABs, neutralizing antibodies; IV, intravenous; CFU, colony forming units.

general clinical condition after treatment. Two recent studies described a total of eleven patients that received nebulized phage therapy targeting *P. aeruginosa* as a monophage (4/11), a phage cocktail (6/11), or two serial doses of monophage (1/11).^{24,25} In most cases, a reduction of *P. aeruginosa* counts in sputum was observed, correlating with improvement in lung function. Another case report of phage nebulization in a CF patient against *P. aeruginosa* combining two phages was recently reported, and although a clinical improvement was observed in the short term, another *P. aeruginosa* strain infected the patient, which was not targeted by the therapeutic phages.¹⁰ While phage resistance was consistently assessed in all these cases, none quantified the immune responses against the phages. Therefore, the immune response to phages after nebulized therapies remains unclear.

Here, we present three nebulized phage therapy compassionate treatments in two CF patients (from La Fe University and Polytechnic Hospital [LFV] and Virgen del Rocío University Hospital [HVR]) with chronic *S. aureus* and/or *P. aeruginosa* lung infection (Table 1). Our results support the safety of the treatments, with no adverse effects described. Interestingly, in one patient, a reduction in cough, expectoration, and need for intravenous antibiotics cycles, with an overall improvement in the quality of life, was observed. However, the analysis of the immune responses suggested the emergence of anti-phage neutralizing antibodies (NABs), which may impair the activity, efficacy, and future use of the phages in these patients. Our results suggest that use of personalized sequential therapies with monophages rather than phage cocktails may be preferred, as phage cocktails could elicit an immune response that limits the range of clinically available phages for future therapies.

RESULTS

Phage suitability, preparation, and safety evaluation

A panel of therapeutic phages from our library was tested on the clinical isolates to determine the efficiency of plating (EOP) of each phage with respect to the amplification strains, selecting phages with EOP ~1 (similar titer in both bacteria). In the case of patient LFV, phage vB_Sau_Mallokai (Mallokai) and LPS-5 were chosen for *S. aureus* and *P. aeruginosa*, respectively. For patient HVR, vB_Pae_10 (phage 10) was selected as the best candidate against *P. aeruginosa*. All three phages were previously used in other patients,^{25,26} providing confidence in their safe use as well suggesting potential therapeutic value. As the annotated phage genomes were not available in public databases when the phage selection was made, the genomic characterization of the three phages was performed. A combination of Illumina and Nanopore sequencing reads resulted in hybrid *de novo* assemblies that aligned with genomes of phages belonging to *Kayvirus* (phage Mallokai), *Pakpunavirus*

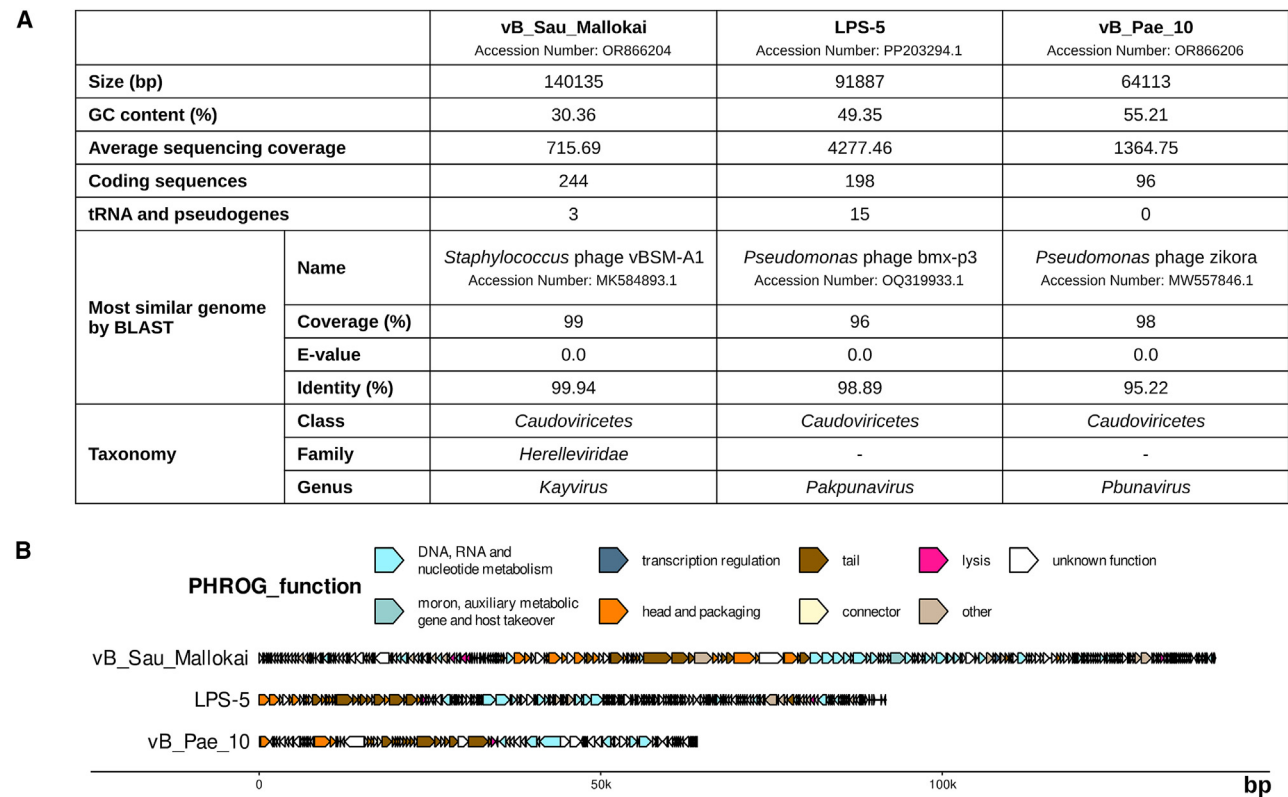


Figure 1. Genomic characterization of the therapeutic phages, Mallokai, LPS-5, and 10

(A) General features of the phage genomes. Genome size, GC percentage, assembled genome coverage, predicted coding sequences, tRNA, and pseudogenes by Pharokka and identification of the closest phage in the nucleotide database through BLASTn analysis are included. (B) Genomic organization and whole-genome comparison of the phage genomes. Each predicted coding sequence is represented by gggenomes³⁰ with an arrow colored based on PHROG's functional category.

(phage LPS-5), and *Pbunavirus* (phage 10). The sequences of the phages are available under the accession numbers GenBank: OR866204, PP203294.1, and ORS866206. Reorganized and corrected genomes based on close relatives lead to genomes ranging from 64,113 to 140,135 pb with a GC content between 30.36% and 55.21% (Figure 1A). Structural and functional annotation with Pharokka identified 96 coding sequences in phage 10, while more than 190 coding sequences were found in both phage LPS-5 and phage Mallokai genomes. Genomic organization of the three phages was similar to other previously described phages,^{27–29} although more than 60% of these identified ORFs remained with unknown function (Figure 1B). Analysis through PhageLeads confirmed the absence of temperate life-style, antimicrobial resistance, and virulence genes.

Phage amplification, production, and vial preparation following an in-house method for the three phages resulted in therapeutic doses containing 2×10^9 plaque-forming units (PFU)/dose for Mallokai and 10^{10} PFU/dose in the case of the *P. aeruginosa* phages LPS-5 and 10. Quality testing of the final vials confirmed the identity of the produced phages, the absence of high amounts of bacterial DNA, low level of endotoxins, and the sterility of the preparations. Stability evaluation showed that phages were stable at 4°C for at least 3 months with no appreciable loss in titer during this time. The effect of the nebulization process by the eFlow rapid device (PARI Pharma) was tested, resulting in 0.30 ± 0.07 log reduction of the phage titer (PFU) for phage 10, 0.42 ± 0.17 log for phage LPS-5, and 0.84 ± 0.31 log for phage Mallokai.

These data, in addition to the medical history and treatment protocol for each patient, were submitted to the *Agencia Española de Medicamentos y Productos Sanitarios* (AEMPS) to request the compassionate use as nebulized treatments. Upon acceptance of the treatments and the follow-up of the patients, the results are described for each case independently.

Case 1: LFV *S. aureus* treatment with phage Mallokai

LFV is a 17-year-old male with CF and chronic infection by methicillin-susceptible *S. aureus* and mucoid *P. aeruginosa*. For *S. aureus* infection, he received oral antibiotics repeatedly over the last 8 years, with several recent cycles of oral linezolid. Due to clinical deterioration as a result of *S. aureus* infection, in July 2023, the patient was treated with a 10-day nebulized administration of phage Mallokai.

Clinical outcome

No adverse effects were observed during phage treatment and subsequent 2 months of follow-up despite a progressive loss of weight from 52.9 kg at day 0 to 46.8 kg 51 days after the phage treatment. The clinical indicators monitored remained almost unchanged and within reference values, with some exceptions including white blood cell (WBC) total count and inflammatory markers (Figure 2A). Neutrophils and monocytes were slightly above the reference ranges on days 0 and 51. Erythrocyte sedimentation rate (ESR) ranged from 63 mm/h to 83 mm/h (always above the reference value), while C-reactive protein (CRP) decreased over the course of the phage therapy from 89.2 mg/L at day 0 to 2.5 mg/L 20 days after completion of the 10-day course of phage administration (corresponding with the lowest bacterial burden). However, on day 51 after phage administration, CRP returned to 100.9 mg/L, its maximum value, in concordance with the increase of bacterial load in sputum. Pulmonary capacity was stable between 52.8% and 54.2% measured as %FEV1/forced vital capacity (FVC). The patient did not report improvement despite the reduction of the bacterial burden.

Bacterial load and susceptibility profile monitoring

Bacterial counts were performed during the follow-up (Figure 2B). Before phage treatment, the *S. aureus* load in the sputum of the patient was higher than 10^6 colony forming units (CFU)/mL, which is the upper limit for bacterial count as measured at the hospital. A progressive decrease in bacterial burden was observed over the course of the phage treatment combined with cotrimoxazole resulting in no growth of *S. aureus* in sputum 2 days after the end of the phage treatment (>5 log reduction). Bacterial count remained around 10^2 – 10^3 CFU/mL during the following week (2–3 log reduction compared to pre-treatment counts) while linezolid was being nebulized. With the next sputum count, performed approximately 1 month after end of the phage treatment and following completion of the antibiotic course, an increase in *S. aureus* was observed with a concentration similar to the pre-treatment state. During this follow-up, the sputum concentration of underlying *P. aeruginosa*, which was minimal (5×10^3 CFU/mL) before the *S. aureus* phage treatment, exhibited a progressive increase. By day 9 post treatment with Mallokai, the concentration of *P. aeruginosa* in sputum (8×10^4 CFU/mL) surpassed that of *S. aureus* (5×10^3 CFU/mL), a trend that persisted throughout the treatment and subsequent weeks until day 42 after treatment initiation, when the latter regained predominance.

Bacterial isolates obtained from sputum over the course of treatment, 20 days and 51 days after the initiation of phage therapy, were analyzed for any change in phage or antibiotic susceptibility (Figure 3). We observed two patterns differentiated by aminoglycosides susceptibility (amikacin, gentamicin, and tobramycin) that could

LFV phage Mallokai

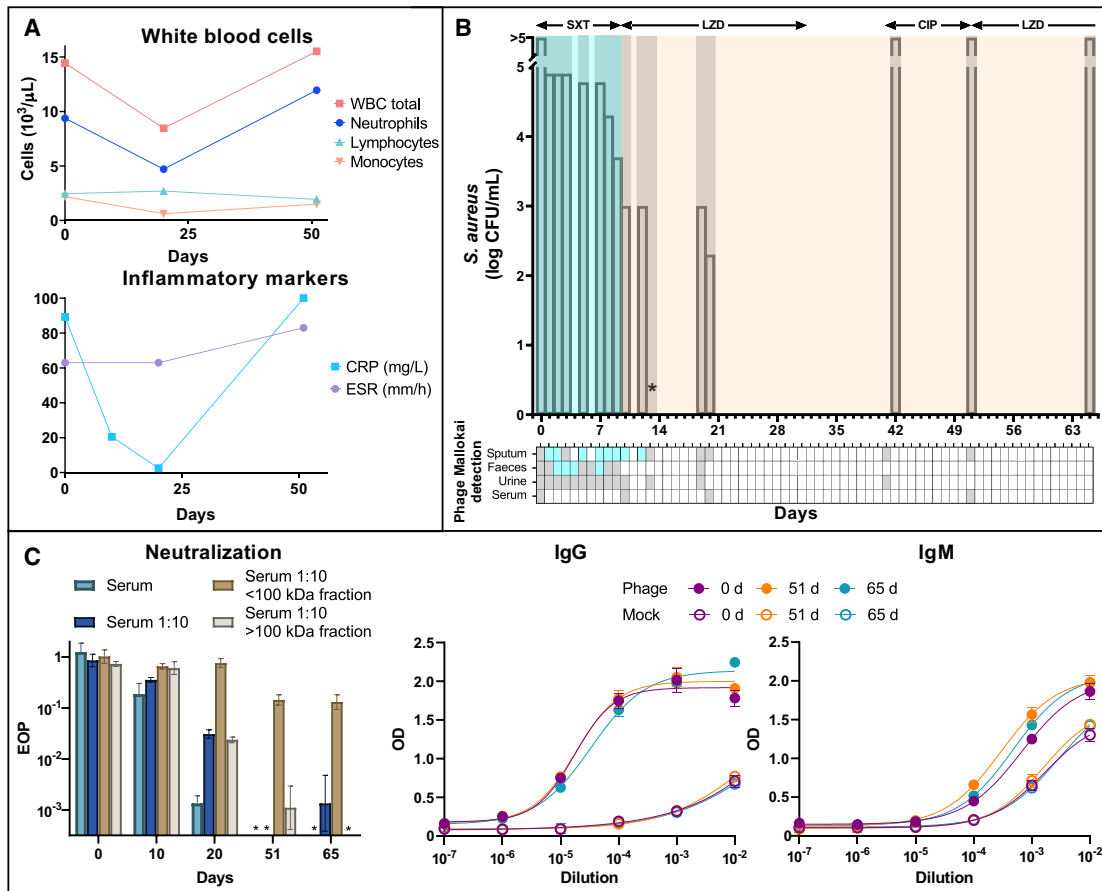


Figure 2. LFV phage Mallokai treatment

(A) Blood biomarkers: white blood cell (WBC) count as 10^3 cells/ μL ; inflammatory markers: C-reactive protein (CRP) measured as mg/L and erythrocyte sedimentation rate (ESR) measured as mm/h.

(B) Bacteria and phage follow-up: log CFU count of *S. aureus* in sputum during and after treatment. (*) indicates that counts exceeded the lower limit of detection. Days in blue correspond to phage treatment. Days in orange indicate the time frame in which neutralization of phage in serum was observed. Antibiotic regime is depicted by arrows on top. Detection of phage Mallokai in patient samples over time is shown on the bottom of each graph. Positive samples are in blue, negative samples are in gray, and not taken or not analyzed samples are in white. SXT, Cotrimoxazole; LZD, Linezolid; CIP, Ciprofloxacin.

(C) Immune response evaluation in serum: neutralizing activity of non-diluted and 1:10 dilution of complete serum and separated fractions below and above 100 kDa were analyzed. Phage titer after serum-phage incubation for 24 h at 25°C was compared to the non-serum control and represented as efficiency of plating (EOP). Three replicates of each dilution were used, and median values with standard deviation are represented. *Indicates that counts exceeded the lower limit of detection. Anti-phage IgG and IgM antibodies are measured by indirect ELISA. For each serum dilution, median and standard deviation of absorbance signal (after background removal) of two biological replicates containing two technical replicates were represented and fitted with a logistic curve.

See also [Figure S1](#).

indicate the presence of different subpopulations of *S. aureus*, all of them susceptible to phage Mallokai (EOP = 1 when comparing all post-treatment isolates with the pre-treatment bacteria).

Phage detection in patient samples

Spot tests and nested PCR were used to check for the presence of phage in a total of 46 samples from sputum ($n = 16$), serum ($n = 5$), urine ($n = 14$), and feces ($n = 11$), taken at different times before, during, and after the phage treatment ([Figure 2A](#)). Mallokai was detected in 4 samples of feces (days 2, 3, 4, and 7 of treatment) and 8 samples of sputum (3 h and days 2, 5, 7, 8, 9, 10, and 12 during

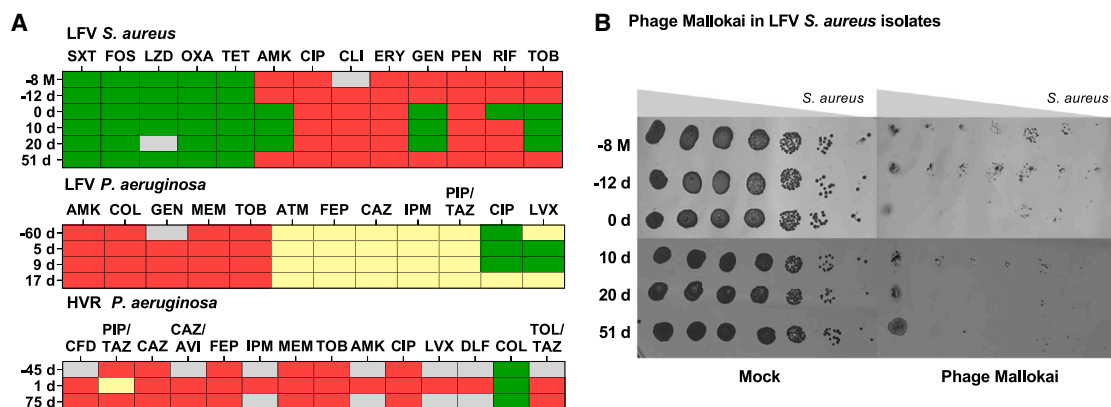


Figure 3. Antibiotic and phage susceptibility of the clinical isolates during the follow-up of the patients

(A) Bacterial susceptibility to antibiotics, colored based on antibiogram result (green, susceptible; red, resistant; yellow, susceptible with increased exposure; gray, not assessed). SXT, cotrimoxazole; FOS, fosfomycin; LZD, linezolid; OXA, oxacillin; TET, tetracycline; AMK, amikacin; CIP, ciprofloxacin; CLI, clindamycin; ERY, erythromycin; GEN, gentamicin; PEN, penicillin; RIF, rifampicin; TOB, tobramycin; COL, colistin; MEM, meropenem; ATM, aztreonam; FEP, cefepime; CAZ, ceftazidime; IPM, imipenem; PIP, piperacillin; TAZ, tazobactam; LVX, levofloxacin; CFD, cefiderocol; AVI, avibactam; DLF, delafloxacin; TOL, ceftolozane. M: months; d: days.

(B) Example of phage Mallokai susceptibility by reverse spot test in LFV *S. aureus* isolates is shown. Similar results are for phage LPS-5 in LFV *P. aeruginosa* and phage 10 in HVR *P. aeruginosa* isolates (data not shown). M: months; d: days.

and after treatment). Phages were not detected in sera or urine at any time point. Phage presence was lastly detected in sputum 3 days after the end of the course of the phage.

Patient immune response evaluation

To analyze the immune response after the treatment, phage neutralization (in sputum and sera) and ELISA were performed (Figure 2C). Phage incubation for 24 h with sputum samples before, during, and after the treatment did not have any effect in phage titer. The same result was observed with patient serum taken before first phage administration. Signs of incipient anti-phage neutralization activity were observed in serum 10 days post administration. The neutralizing effect increased over time, reaching >3 log PFU reduction (limit of detection), when incubated with serum from 51 days post administration. This level of neutralization was maintained in sera until the last sample was analyzed (65 days after treatment).

To identify the cause of the neutralization, sera were separated by centrifugation in two fractions, >100 kDa, which should retain most antibodies contained in the sera, and <100 kDa for smaller molecules. Neutralization activity occurred in the >100 kDa fraction. Neutralization activity was not observed after incubation with an unrelated, control phage, suggesting the specificity of the neutralization. Simultaneously, neutralization assays using heat-inactivated sera were performed to test the activity of the complement. No differences were observed using either not treated or heat-inactivated sera. In addition, ELISA was performed to detect immunoglobulin G (IgG) and IgM antibodies in sera from days 0, 51, and 65. Despite neutralizing capacity of sera increasing over the time, anti-phage Mallokai IgG and IgM abundances were equivalent in a non-neutralizing pre-treatment sample and two neutralizing samples from days 51 and 65 after treatment.

Phage-antibiotic interactions

Since the phage treatment was done in combination with antibiotics, potential interactions between the phage and the antibiotics were tested. At all the multiplicities of

infection (MOIs) tested (0.001–10), the phage itself was able to completely prevent bacterial growth, where cotrimoxazole only suppressed bacterial growth at the highest concentration tested (0.008/0.4 mg/mL). In the case of linezolid, 0.01 mg/mL eradicated the bacteria effectively, while 0.001 mg/mL delayed bacterial growth for 7 h, and 0.0001 mg/mL had no impact in the bacterial growth. Although synergy could not be evaluated as the phage alone was sufficient to effectively kill the bacteria at all the MOIs tested, none of the combinations of phage and antibiotic showed antagonism, maintaining the same effect of phage alone (Figure S1).

Case 2: LFV *P. aeruginosa* treatment with phage LPS-5

As previously mentioned, LFV is chronically infected by *S. aureus* and *P. aeruginosa*. Alternating nebulized colistin and tobramycin was the treatment for *P. aeruginosa* infection. The bacterial burden of *P. aeruginosa* was significantly lower than the predominant infection of *S. aureus* in this patient at the onset of treatment (i.e., Case 1). However, due to the increase of *P. aeruginosa* after treatment with phage Mallokai, in September 2023, the patient was treated with a 10-day course of phage LPS-5.

Clinical outcome

LFV remained stable during the treatment, with no adverse effects associated with phage administration described. Clinical markers were constant, with some outside the recommended values (Figure 4A), including CRP ranging between 8.8 and 12 mg/L, the monocytes on days 0 and 28, and the eosinophils and platelets on day 0 (data not shown). %FEV/FVC increased to 60.66% 1 week after phage LPS-5 administration, showing improvement when compared to the measurement recorded 1 month before phage treatment (53.92%). The weight of the patient remained stable, being 47.4 kg and 46.8 kg, respectively.

Bacterial load and susceptibility profile monitoring

Bacterial counts were performed to evaluate the efficacy of phage LPS-5 (Figure 4B). Unfortunately, samples taken on days 1, 14, and 17 post treatment were not analyzed on the same day and were kept at 4°C. *P. aeruginosa* can grow at this temperature, which may lead to an overestimation of results. However, the concentration in sputum was relatively low, between 4×10^3 and 1×10^5 CFU/mL through the treatment and follow-up, and no effects of the phage and the antibiotic treatments were observed in terms of bacterial reduction. It is worth mentioning that on day 7, no bacterial growth was observed, but sputum taken 2 days later showed the presence of bacteria at the concentration previously observed. Bacterial isolates obtained during and after the treatment showed no changes either in antibiotic (Figure 3A) or phage sensitivity profile (maintenance of EOP with respect to the pre-treatment isolate).

Phage detection in patient samples

The presence of phage LPS-5 was tested by nested PCR in 8 samples from sputum taken at different times before, during, and after the phage treatment (Figure 4B). Phages were detected in sputum in all samples, from the day of treatment initiation to day 17, the last available sample (8 days after the last nebulization).

Patient immune response evaluation

Immune responses were analyzed during treatment. Phage neutralization assays were performed with all the available sputum and sera (Figure 4C). The results showed no effects after 24-h incubation with phage LPS-5 in any sputum samples. In sera, a slight reduction (between 0.3 and 0.6 log PFU) was observed 17 days after treatment initiation and maintained on days 28 and 45. However, this effect

LFV phage LPS-5

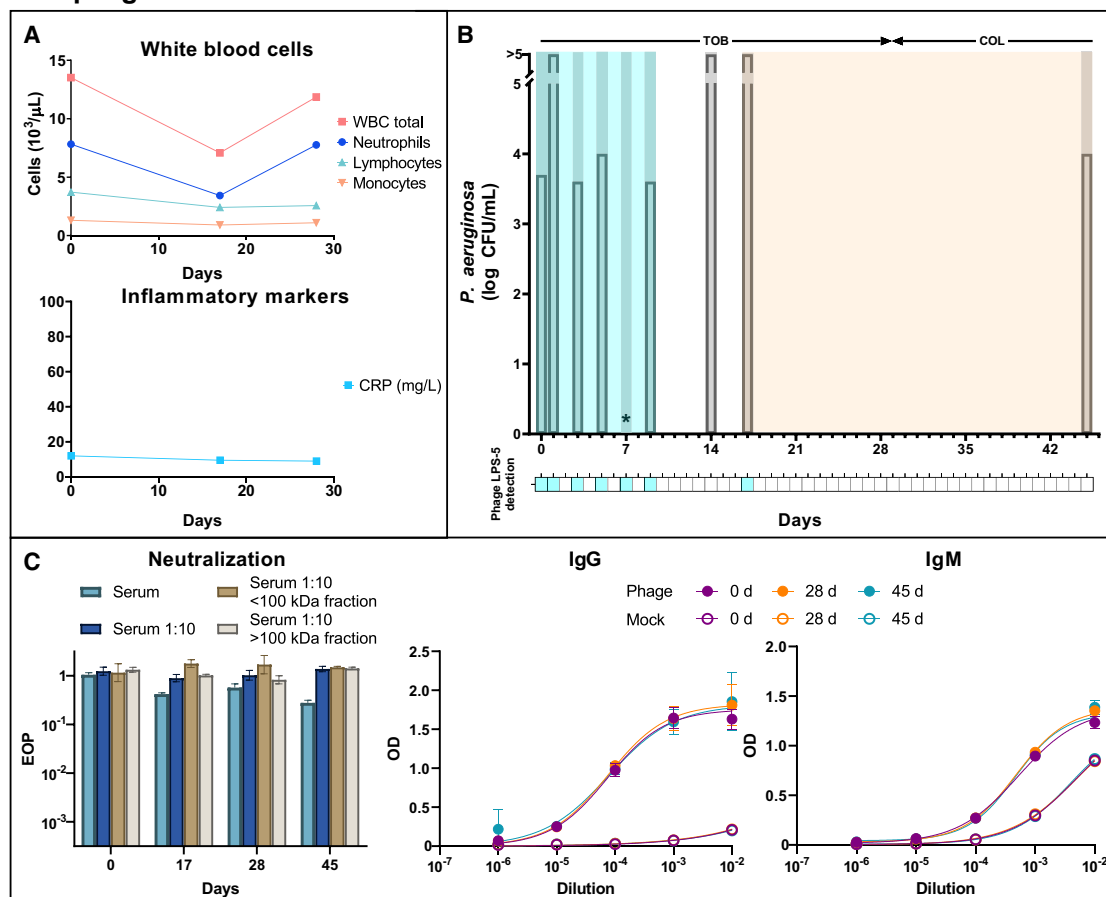


Figure 4. LFV phage LPS-5 treatment

(A) Blood biomarkers: white blood cell (WBC) count as 10^3 cells/ μL ; inflammatory markers: C-reactive protein (CRP) measured as mg/L.

(B) Bacterial and phage follow-up: log CFU count of *P. aeruginosa* in sputum during and after treatment. (*) indicates that counts exceeded the lower limit of detection. Days in blue correspond to phage treatment. Days in orange indicate the time frame in which neutralization of phage LPS-5 in serum was observed. Antibiotic regime is depicted by arrows on top. Detection of phage in patient samples over time is shown on the bottom of each graph. Positive samples are in blue, negative samples are in gray, and not taken or not analyzed samples are in white. TOB, Tobramycin; COL, Colistin.

(C) Immune response evaluation in serum: neutralizing activity of non-diluted and 1:10 dilution of complete serum and separated fractions below and above 100 kDa were analyzed. Phage titer after serum-phage incubation for 24 h at 25°C was compared to the non-serum control and represented as EOP. Three replicates of each dilution were used, and median values with standard deviation are represented. *Indicates that counts exceeded the lower limit of detection. Anti-phage IgG and IgM antibodies are measured by indirect ELISA. For each serum dilution, median and standard deviation of absorbance signal (after background removal) of two biological replicates containing two technical replicates were represented and fitted with a logistic curve.

disappeared when the sera was diluted 1:10, suggesting the presence of NAb at low concentration. Complement system inactivation by heat and 100-kDa cutoff separation supported this hypothesis. A phage control showed no reduction in titer in any sample. ELISA assays of sera (days 0, 28, and 45 post administration) showed no changes in IgG or IgM anti-phage LPS-5 levels (Figure 4C).

Case 3: HVR *P. aeruginosa* treatment with phage 10

HVR is 37-year-old female with CF and pulmonary infection with *P. aeruginosa* treated with nebulized colistin combined with aztreonam or levofloxacin, requiring intravenous antibiotic therapy every 1 or 2 months. In August 2023, the patient was treated with a 10-day nebulized course of phage 10.

HVR phage 10

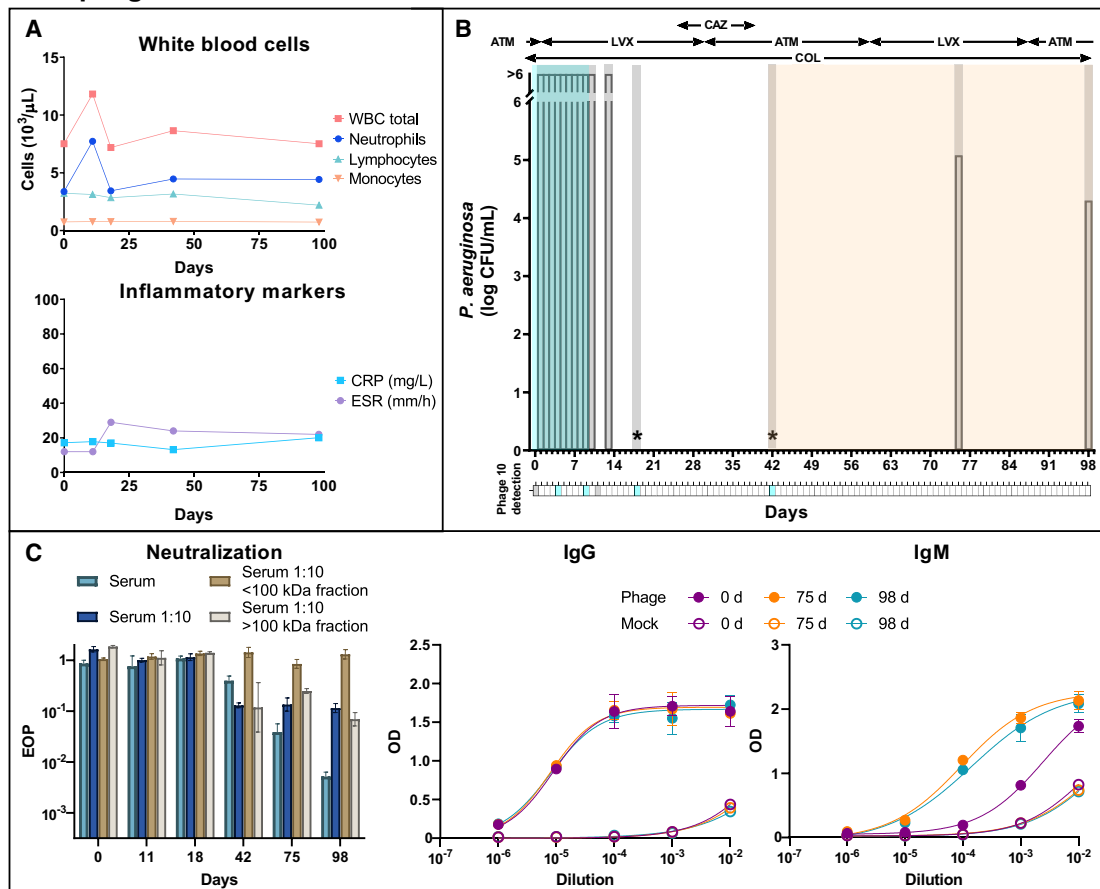


Figure 5. HVR phage 10 treatment

(A) Blood biomarkers: white blood cell (WBC) count as 10^3 cells/ μL ; inflammatory markers: C-reactive protein (CRP) measured as mg/L and erythrocyte sedimentation rate (ESR) measured as mm/h.

(B) Bacterial and phage follow-up: log CFU count of *P. aeruginosa* in sputum during and after treatment. (*) indicates that counts exceeded the lower limit of detection. Days in blue correspond to phage treatment. Days in orange indicate the time frame in which neutralization of phage in serum was observed. Antibiotic regime is depicted by arrows on top. Detection of phage 10 in patient samples over time is shown on the bottom of each graph. Positive samples are in blue, negative samples are in gray, and not taken or not analyzed samples are in white. ATM, Aztreonam; LVX; Levofloxacin; CAZ, Ceftazidime; COL, Colistin.

(C) Immune response evaluation in serum: neutralizing activity of non-diluted and 1:10 dilution of complete serum and separated fractions below and above 100 kDa were analyzed. Phage titer after serum-phage incubation for 24 h at 25°C was compared to the non-serum control and represented as EOP. Three replicates of each dilution were used, and median values with standard deviation are represented. *Indicates that counts exceeded the lower limit of detection. Anti-phage IgG and IgM antibodies are measured by indirect ELISA. For each serum dilution, median and standard deviation of absorbance signal (after background removal) of two biological replicates containing two technical replicates were represented and fitted with a logistic curve.

Clinical outcome

The patient reported a reduction of coughs and mucus production 48 h after therapy initiation and overall well-being improvement that continuously increased over the course of the nebulized phage therapy and that prevails to this day (April 2024). Moreover, the patient's need for IV therapy to control bacterial exacerbations disappeared. No adverse effects were described during treatment and subsequent follow-up, with most clinical indicators being constant during all the time. CRP was above the reference values throughout all the monitoring and ESR also from day 18 onward. There was a particular peak in neutrophils on day 11, while the rest of blood cell counts were stable (Figure 5A). %FEV1/FVC was fluctuating

between 47.18% 3 days before and 48.54% 98 days after the start of phage treatment. Weight was stable around 60.8 kg.

Bacterial load and susceptibility profile monitoring

Bacterial counts in sputum remained stable and above the limit of detection (10^6 CFU/mL) during phage treatment (Figure 5B). On days 18 and 42 after phage treatment initiation, a >6 log CFU reduction was observed, since no *P. aeruginosa* was detected in sputum. At days 75 and 98, *P. aeruginosa* was again isolated, with concentrations of 1.2×10^5 CFU/mL and 2×10^4 CFU/mL respectively, both lower than the pre-treatment concentration. The antibiotic susceptibility of the bacterial isolates from days 75 and 98 remained unchanged (Figure 3A). Phage sensitivity was tested on an isolate obtained 112 days after phage initiation, remaining susceptible with an EOP of 1 in comparison to the pre-treatment isolate.

Phage detection in patient samples

Nested PCR was used to check for the presence of phage 10 in a total of 7 samples from sputum taken at different times before, during, and after the phage treatment (Figure 5A). As previously observed for the other phages, phage 10 was detected in sputum samples taken over the course of the treatment (days 4 and 9). Additionally, phage presence was obtained in two sputum samples taken on days 18 and 42 days after phage treatment initiation, more than 1 month after the phage was lastly nebulized, suggesting the replication of the phage at the site of infection. No further samples were available to test phage presence later than day 42.

Patient immune response evaluation

Neutralizing assays in sputum samples showed no reduction in phage titer after their co-incubation for 24 h (Figure 5C). Sera taken before treatment and 11 and 18 days after the first phage nebulization had no impact on reducing titer of phage 10. In contrast, on day 42, a slight decrease of titer (0.5 log PFU) was observed, increasing on days 75 and 98, with a 1.5 and >2 log reduction in PFU, respectively. These effects were maintained with the 1:10 dilution of sera and the fraction >100 kDa but disappeared in the fraction <100 kDa, suggesting the effect of NAb. No reduction was observed in any of the samples for a control phage. The neutralizing effect was in concordance with an increased anti-phage 10 IgM antibody level when comparing the serum before treatment and the sera 75 and 98 days after phage administration (Figure 5C). No changes for IgG levels before and after treatment were observed for the same samples.

DISCUSSION

Here, we present three phage nebulized treatments in two CF patients. Both patients suffer from chronic infection of *S. aureus* and/or *P. aeruginosa*, with long-term antibiotic treatment, significantly impacting their quality of life without obtaining a sustained clinical improvement. Under these circumstances, phage therapy was proposed as compassionate use, maintaining the standard-of-care antibiotics during the treatment, as previously reported.²⁵ Following an in-house protocol at the Institute for Integrative Systems Biology (Valencia, Spain), we generated the first phage preparations approved by the AEMPS in Spain. Due to the lung infections in the CF patients, nebulization was the method of administration, delivering the phages directly to the infection site.³¹ The eFlow rapid device was the nebulizer selected, given the availability and routine use by CF patients in Spain,^{32,33} offering the possibility of treatment without the need for hospitalization in a comfortable procedure. Moreover, the stability of the three phages through the nebulizer membrane was tested to avoid the risk of loss of phage viability, with results that support the use of these devices for nebulized phage therapy, as previously reported.^{34,35}

10-day phage treatments were well tolerated without adverse events. A slight weight loss in one patient was observed that, although temporally correlated, could not be clearly associated with phage treatment. In addition, the clinical outcome was favorable for a patient, with a drop in the *P. aeruginosa* count that resulted in a reduction in cough and expectoration, with an overall well-being improvement. This progress started over the course of the treatment and persisted to the present day with no need of intravenous antibiotic cycles, which were administered recurrently to control bacterial exacerbations. This positive outcome was observed despite no complete eradication of bacteria, as previously described in other *P. aeruginosa* phage treatments.³⁶ In the other patient, chronically co-infected with *S. aureus* and *P. aeruginosa* and subjected to two serial treatments with monophages, no improvement was observed. A possible explanation for the result, despite the reduction in CFU of the phage-target bacterium, could be the population dynamics between both bacteria. Supporting this, the significant reduction in *S. aureus* resulted in an increase in *P. aeruginosa* counts, which could explain the patient outcome, suggesting that in cases of multi-species infections, a combination of two simultaneously monophage treatments could be preferable in order to target both bacteria. On the other hand, the treatment directed to *P. aeruginosa* showed no significant differences.

To better understand the results, phage resistance emergence and immune response after each treatment were explored. Emergence of phage resistance, which is seen as a major issue for phage therapy, did not occur in any of the cases, which is not uncommon, as it occurred in previous treatments.^{10,11} However, sequencing of bacteria and phages could be interesting to understand *in vivo* coevolution. Phage immunogenicity has been described previously,⁹ although neutralizing activity is usually more associated with invasive routes of administration such as intravenous or intrathecal¹⁰ rather than oral or nebulized therapies. However, our results revealed that nebulized phage therapies can elicit specific immune responses, mainly through NAb. This is supported by the specificity of the neutralization to the nebulized phages, with no effects on control phages. In addition, the neutralization activity was detected in the >100 kDa sera fraction, and the complement system action was discarded by heat inactivation, supporting the emergence of anti-phage-specific antibodies. The occurrence of neutralization correlated with halted bacterial reduction in two of the cases, suggesting that specific anti-phage antibodies might impair the effectiveness of the phages as previously described.³⁷ However, this might not prevent favorable clinical responses as it occurred in the case of HVR. Despite detecting neutralizing activity for all three phages, only one case resulted in an increase of anti-phage IgM. This is in agreement with prior studies, showing that low to medium neutralizing activity did not always correlate with an increment of IgG or IgM levels by ELISA assays with significant changes observed when neutralization effect was very high.³⁸ The results observed here support the activation of the immune system after localized non-invasive therapies, despite the controlled delivery of the phages to the lungs.

To the best of our knowledge, we report here the first emergence of NAb after phage nebulized treatments. NABs were detected in all three cases following the 10-day nebulized therapy. Indeed, bacterial regrowth was observed after the detection of neutralizing activity in all cases. However, the impact of the anti-phage immune response on the clinical outcome is variable, since previous cases resulted in both successful^{39,40} and failed³⁷ treatments, as seen here. Moreover, immune responses ranged from 10 to 42 days, highlighting the need for large monitoring periods even after short-term phage therapies. Interestingly, our

patients showed detectable levels of binding IgM and IgG against the nebulized phages before their administration, as reported previously.^{37–39} This could be a consequence of natural exposure to other related phages during life that may cause immune cross-reactivity with the administered phages.³⁹ Thus, evaluation of the immunogenicity of the therapeutic phages before their administration might prevent negative outcomes, even when non-invasive routes are chosen. However, further studies are required to better understand the phage determinants that trigger immune responses.

Phage therapy is considered a safe and promising treatment for MDR bacterial infections. However, clinical trials are mandatory to implement phages as standard-of-care treatments. The emergence of NAb after nebulized treatments, reported here for the first time, highlights the importance of implementing standardized monitoring protocols. These should include assessment of phage immunogenicity and patient immune status even before initiation of the treatment, regardless of the route of administration. Furthermore, our results strongly recommend phages as personalized therapies, avoiding ready-to-use phage cocktails to prevent unnecessary exposition to phages, which could elicit NAb, reducing the repertoire of effective phages for future treatments.

Limitations of the study

Given the non-trial/uncontrolled nature of the study with a limited number of cases ($n = 3$), one must be cautious in the interpretation of the results. NAb presence after phage nebulization was observed in the three treatments. However, these findings must be further evaluated through clinical trials, since patient immune state could have an impact on the development of NAb. All three treatments were combined with antibiotics, limiting the assessment of the contribution of phages to patient outcome.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [RESOURCE AVAILABILITY](#)
 - Lead contact
 - Materials availability
 - Data and code availability
- [EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS](#)
 - Patients' data
 - Bacterial strains and clinical isolates
 - Therapeutic phages
 - Compassionate treatment approval
- [METHOD DETAILS](#)
 - Study design
 - Bioinformatics analysis of the therapeutic phages
 - Phage preparation, safety, and stability
 - Phage administration and treatment monitoring
 - Phage titration by spot test assays
 - Phage molecular detection
 - Bacterial count and susceptibility profile
 - Phage neutralization assays
 - Immunological assays for anti-phage antibodies detection
 - Evaluation of phage-antibiotic interactions
- [QUANTIFICATION AND STATISTICAL ANALYSIS](#)

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.medj.2024.05.017>.

ACKNOWLEDGMENTS

The authors would like to thank the patients and the AEMPS for accepting the compassionate use of the phages; the hospital staff in the departments of microbiology and pharmacy for their technical assistance; PARI Pharma GmbH for kindly providing the eFlow rapid device; and Carlos Sellés, Miranda Tomás, and Irene Cantalops for their technical support. The project was funded by crowdfunding in collaboration with the Spanish Cystic Fibrosis Foundation, Project SEJIGENT/2021/014 (Conselleria d'Innovació, Universitats, Ciència i Societat Digital; Generalitat Valenciana), Project PID2020-112835RA-I00 (Spanish Ministry of Science, Innovation and Universities), and Ramón y Cajal contract RYC2019-028015-I (Spanish Ministry of Research and Innovation).

AUTHOR CONTRIBUTIONS

Design and interpretation of the experiments: M.B.-G., M.P.-F., B.K.C., and P.D.-C. Performed the experiments: M.B.-G., M.P.-F., and A.G.-B. Clinical investigators: P.E.T., S.P.-T., L.C.-H., and E.Q.-G. Funding acquisition: P.D.-C. Supervision: P.D.-C. Statistical analysis and writing – original draft: M.B.-G. and M.P.-F. Writing – review and editing: B.K.C., P.E.T., A.G.-B., S.P.-T., L.C.-H., E.Q.-G., and P.D.-C. Unrestricted access to all data: M.B.-G., M.P.-F., and P.D.-C. All authors read and approved the final article and take responsibility for its content.

DECLARATION OF INTERESTS

P.D.-C. is cofounder of Evolving Therapeutics SL and a member of its scientific advisory board, and P.E.T. is cofounder of Felix Biotechnology, Inc.

Received: February 28, 2024

Revised: April 2, 2024

Accepted: May 30, 2024

Published: June 24, 2024

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Goat anti-Human IgM (Heavy chain) Cross-Adsorbed Secondary Antibody, HRP	ThermoFisher Scientific	Cat#A18841; RRID: AB_2535618
Goat anti-Human IgG Fc Cross-Adsorbed Secondary Antibody, HRP	ThermoFisher Scientific	Cat#A18823; RRID: AB_2535600
Bacterial and virus strains		
<i>Staphylococcus aureus</i> subsp. <i>aureus</i> Rosenbach	Spanish Type Culture Collection	CECT 239
<i>Pseudomonas aeruginosa</i> -Schroeter-Migula	Spanish Type Culture Collection	CECT 4122
Phage Mallokai	Center for Phage Biology & Therapy, Yale University	N/A
Phage LPS-5	Center for Phage Biology & Therapy, Yale University	N/A
Phage 10	Center for Phage Biology & Therapy, Yale University	N/A
LFV clinical isolates	This paper	N/A
HVR clinical isolates	This paper	N/A
Biological samples		
LFV and HVR serum	This paper	N/A
LFV and HVR sputum	This paper	N/A
LFV urine	This paper	N/A
LFV feces	This paper	N/A
Chemicals, peptides, and recombinant proteins		
Vegetal Trypto-Casein Soy Broth	Sigma-Aldrich	Cat#1.00550
Endotoxin-Free Dulbecco's PBS (1X) (w/o Ca++ & Mg++)	Sigma-Aldrich	Cat#TMS-012
Magnesium sulfate solution 1M	ThermoFisher Scientific	Cat#J61030-AK
Tween 20	Sigma-Aldrich	Cat#P1379
Bovine Serum Albumin	Sigma-Aldrich	Cat#A9647
Sodium bicarbonate	Sigma-Aldrich	Cat#S5761
Sodium carbonate (anhydrous)	ThermoFisher Scientific	Cat#S/2880/53
3,3',5,5'-tetramethylbenzidine substrate	Sigma-Aldrich	Cat#T0440
Sulfuric acid	Sigma-Aldrich	Cat#258105
Cotrimoxazole	La Fe University and Polytechnic Hospital	N/A
Linezolid	La Fe University and Polytechnic Hospital	N/A
Critical commercial assays		
Ultrafilter Concentrating Pipette Tips	Innova prep	Cat#CC08003-10
PBS FluidPrep Elution Fluid	Innova prep	Cat#HC08000
Endotrap HD	Biovender	Cat#LET0010
Endotoxin-content assay	Laboratorio Echevarne	N/A
Sterility assay	Laboratorio Echevarne	N/A
Maxwell RSC PureFood GMO and Authentication Kit	Promega Biotech	Cat#AS1600
Illumina sequencing 150 bp paired-end	Universitat de València	N/A
Minion flow cell R9.4.1.1	VWR	Cat#OXNTFLO-MIN106D
Rapid Barcoding Kit 96	VWR	Cat#SQK-RBK110.96
Deposited data		
Phage Mallokai genome	This paper	GenBank Accession Number OR866204
Phage LPS-5 genome	Chan et al. ²⁵	GenBank Accession Number PP203294.1
Phage 10 genome	This paper	GenBank Accession Number ORS866206

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
GraphPad Prism	GraphPad Software	www.graphpad.com
Unicycler	Wick et al. ⁴¹	https://github.com/rwick/Unicycler
BLAST	Altschup et al. ⁴²	https://blast.ncbi.nlm.nih.gov/Blast.cgi
Pilon	Walker et al. ⁴³	https://github.com/broadinstitute/pilon
Pharokka	Bouras et al. ⁴⁴	https://github.com/gbouras13/pharokka
gggenomes	Ankenbrand and van Adrichem ³⁰	https://github.com/thackl/gggenomes
PhageLeads	Yukgehnash et al. ⁴⁵	https://phageleads.dk/
Other		
eFlow@rapid device	PARI Pharma GmbH	Cat#678G1020

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Pilar Domingo-Calap (pilar.domingo@uv.es).

Materials availability

This study did not generate new unique reagents.

Data and code availability

Phage genomic sequencing data have been deposited at GenBank and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#). Primers sequence for specific phage detection can be found at [Table S1](#). This paper does not report original code. Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Patients' data

Patient LFV (La Fe Valencia) is a 17-year-old male CF patient homozygous for the W1089X mutation, non-eligible for modulator therapies, with chronic infection by methicillin-susceptible *S. aureus* and mucoid *P. aeruginosa* since he was 4 and 5 years old, respectively. For *S. aureus* infection, he received oral antibiotics repeatedly over the last 8 years, with several recent cycles of oral linezolid, while alternating nebulized colistin and tobramycin was the treatment for *P. aeruginosa* infection. In recent years, there was progressive deterioration in lung function (especially in the last 3 years) with much greater involvement of the left lower lobe. Suffering from recurrent febrile episodes during the last 6 years, he required several cycles of intravenous antibiotics at home or during hospitalization. In October 2022, the possibility of using phage therapy for treating *S. aureus* and *P. aeruginosa* infections was raised at the request of the family to La Fe University and Polytechnic Hospital (Valencia, Spain), after receiving information from the Spanish Cystic Fibrosis Foundation.

Patient HVR (Virgen del Rocío University Hospital) is a 37-year-old female CF patient with G542X/R1066C mutations, non-eligible for modulator therapies, suffering from bronchiectasis and chronic infection of *P. aeruginosa*. Bacterial infection was treated with nebulized colistin combined with aztreonam or levofloxacin requiring intravenous antibiotic therapy every 1 or 2 months. Radiographic imaging multiple abnormal dilations of the bronchi, especially in the central area of the lungs, with a greater impact in the middle and upper regions. Since September 2022, the patient has been on the active transplant list. In December 2022, responsible physician

at the Virgen del Rocío University Hospital (Sevilla, Spain) contacted the researchers about the possibility of using phage therapy to treat *P. aeruginosa* infection.

Bacterial strains and clinical isolates

The reference strains, *S. aureus* subsp. *aureus* Rosenbach (SA239) and *P. aeruginosa* -Schroeter-Migula (PA4122) were obtained from the Spanish Type Culture Collection (CECT). Clinical isolates from patients were obtained from La Fe University and Polytechnic Hospital (LFV isolates) and Virgen del Rocío University Hospital (HVR isolates).

SA239, PA4122 and patient bacterial isolates were grown overnight in lysogeny broth (LB), supplemented with 5 μ M CaCl₂, and shaking at 37°C. For phage vials preparation, bacteria were grown in vegetal Trypto-Casein Soy Broth (TSB) with shaking at 37°C.

Therapeutic phages

S. aureus phage vB_Sau_Mallokai (phage Mallokai) and *P. aeruginosa* phages LPS-5 (phage LPS-5) and vB_Pae_10 (phage 10) were kindly provided by Drs. Chan and Turner, previously used in phage therapy in the USA.^{25,26}

Compassionate treatment approval

In the absence of a regulation for phage therapy in Europe, the AEMPS is responsible for the validation of compassionate phage therapy in Spain. Institutional Review Boards of La Fe University and Polytechnic Hospital (Valencia, Spain) and Virgen del Rocío University Hospital (Sevilla, Spain) gave ethical approval for this work before the initiation of the treatments. All necessary patient consent was obtained and the appropriate institutional forms were archived. All patient information present has been de-identified.

METHOD DETAILS

Study design

We described three compassionate treatments of monophage therapies in two CF patients with chronic pulmonary infection by *S. aureus* and/or *P. aeruginosa*. Based on our previous experience, phages were administered by nebulization for 10 days in combination with the regular antibiotic regime of each patient.

For eligibility to compassionate phage therapy, microbiologic documentation of MDR bacteria in at least 3 positive cultures over time (more than a year), documenting resistance, evidence of clinical failure and absence of available treatments are mandatory. Bacterial isolates for each patient are tested against a collection of therapeutic phages selecting phages with an EOP ranging from 0.1 to 1 (with 1 being the same titer) in the patient's isolates in comparison to the reference strain. When available, at least three isolates were tested for phage susceptibility before treatment. In addition, an in-house protocol for production and preparation of phage vials at the Institute for Integrative Systems Biology (University of Valencia-CSIC, Spain) was approved by the AEMPS.

Bioinformatics analysis of the therapeutic phages

High-titre lysates of phages were used for DNA extraction and subsequent sequencing using Illumina and Nanopore technology. The *de novo* genome assembly was performed using Unicycler⁴¹ followed by the discard of host contamination sequences and BLAST⁴² analysis of the remaining contig against nucleotide database to search for close relatives. To facilitate whole-genome comparison,

beginning of related phages was set as the start of each genome. Reorganized genomes were corrected using Pilon⁴³ and Illumina sequencing information leading to more reliable genomes that were used to perform the structural and functional annotation with Pharokka.⁴⁴ Coding sequences were grouped by PHROG functional category⁴⁶ and represented with gggenomes.³⁰ Lastly, PhageLeads⁴⁵ analysis was used to ensure suitability of the phages Mallokai, LPS-5 and 10 for therapy.

Phage preparation, safety, and stability

Phages were amplified using the reference strains in 900 mL of non-animal origin TSB. After complete lysis of the culture, lysates were centrifuged and filtered through 0.45 nm and 0.22 nm pore filters to remove bacterial debris. Filtered lysates were concentrated to minimum volume using FluidPrep Concentrating Pipette Select with Ultrafilter Concentrating Pipette Tips and eluted in PBS FluidPrep Elution Fluid (Innova Prep, Drexel, MO). Concentrated phages were diluted in PBS 1X + MgSO₄ 10 mM. For phages LPS-5 and 10, endotoxins were removed using EndoTrap HD. Then, three rounds of dialysis were performed and phage solutions were diluted in PBS 1X + MgSO₄ 10 mM to produce phage vials. Each vial contained 2×10^9 particle forming units (PFUs) for phage Mallokai, and 1×10^{10} PFU in the case of phage LPS-5 and 10.

Each production batch was sent to an external reference laboratory (Laboratorio Echevarne, S.A.) to assess sterility according to European Pharmacopoeia XI Ed. - 2.6.1 (Filtration Technique) and endotoxin content following the European Pharmacopoeia XI Ed. (2.6.14-method D) and the American Pharmacopoeia USP (85 - Chromogenic Technique). Additionally, one vial of each batch was used for total DNA extraction to confirm phage identity and evaluate traces of bacterial DNA.

Phage stability in vials at 4°C was examined monthly during the first 3 months and before the treatment by checking the phage titer in SA239 and PA4122. Additionally, phage nebulization-induced titer loss was evaluated *in vitro* to ensure the efficacy of the chosen delivery device. Briefly, three replicates of phage vials were nebulized using an eFlow®rapid device (PARI Pharma GmbH) connected to a collector. Phage titer before and after nebulization was compared to assess phage stability through the vibrating membrane.

Phage administration and treatment monitoring

Nebulized administration of the phages was done daily using an eFlow®rapid device for 10 days. All phage treatments were combined with the regular antibiotic regime of each patient. LFV treatment of *S. aureus* was done using phage Mallokai combined with cotrimoxazole followed by linezolid for two weeks. For phage LPS-5 treatment, colistin and tobramycin were alternately taken for one month. In the case of HVR, phage 10 was combined with colistin and either aztreonam or levofloxacin which the patient alternates in 28 days cycles.

In all cases, first dose of phage therapy was administered under medical supervision. Blood, sputum, urine and feces samples were taken before and after phage therapy initiation on that day, periodically over the course of the treatment, and after the end of phage therapy. When possible, the presence of phage was evaluated in these samples. Sputum samples were also used to measure bacterial load and obtain bacterial isolates to determine possible changes in phage and antibiotic susceptibilities. Additionally, blood samples were used to monitor clinical indicators (blood cells count, platelets and hemoglobin, liver function markers and inflammatory markers) and centrifuged to obtain the sera. The immune responses against the nebulized

phage were assessed in both sera and sputum samples. Other indicators such as weight and pulmonary function were evaluated periodically.

Phage titration by spot test assays

Phage efficiency in the bacterial strains was tested using 1 μ L droplets of phage dilutions spotted over bacterial monolayers as previously described.⁴⁷ Efficiency of plating (EOP)⁴⁸ was calculated to test phage therapeutic suitability (plating in parallel the reference strains SA239 or PA4122 and the clinical isolates) and to evaluate changes in phage susceptibility (plating in parallel both pre and post treatment isolates). Alternatively, reverse spot test approach was done using phage monolayers with 1 μ L droplets of bacterial dilutions to compare phage susceptibility of all different isolates to the same phage exposure.

Additionally, similar assays were used for phage detection in the samples of the patients, by adding a 1 μ L droplet of each sample in triplicate over an SA239 or PA4122 lawn in semi-solid agar. Samples with phage detected using this technique were considered positive for phage presence.

Phage molecular detection

Phage presence in patient samples was also screened by nested PCR using primers listed in [Table S1](#). Firstly, viral DNA from 3 replicates of each sample were extracted using Maxwell RSC PureFood GMO and Authentication Kit (Promega Biotech Ibérica S.L.). Samples with phage detected by nested PCR in at least two replicates were considered positive for phage presence.

Bacterial count and susceptibility profile

Bacterial counts from patient sputum samples were measured by quantitative seeding at the microbiology departments from La Fe University and Polytechnic Hospital (Valencia, Spain) and Virgen del Rocío University Hospital (Sevilla, Spain). When possible, patient isolates were screened for antibiotic-resistance profile and phage susceptibility in comparison to the pre-treatment isolates was tested using the spot test methodology described above.

Phage neutralization assays

Phage-specific neutralizing capacity of sputum and sera were evaluated through neutralization assays as described.³⁷ A preliminary assay was performed to test all sputum and sera obtained from the patients. Briefly, 90 μ L of phage solution in PBS 1X was mixed with 10 μ L of each serum and sputum in triplicate, and incubated at 25°C for 24 h. After incubation, phage titer in all the mixtures was analyzed for any decrease in comparison to the non-serum control. To determine the fraction responsible for the neutralization, an 1:10 dilution in PBS 1X of sera were centrifuged at 4000g using a 100 kDa cut-off to analyze the neutralizing capacity of total sera and their fractions above and below 100 kDa. After centrifugation, the fraction <100 kDa was collected, and the fraction retained in the 100 kDa cut-off filter was resuspended in a minimum volume. Neutralizing effect of the samples was evaluated as described above. In parallel, sera were heat inactivated at 56°C for 30 min to discard neutralization due to the complement system.⁴⁹

Immunological assays for anti-phage antibodies detection

Selected sera taken before and after neutralization detection were screened for the presence of anti-phage IgM and IgG antibodies using an indirect ELISA technique as previously described³⁷ with minor modifications. In summary, three replicates of 10-fold dilutions of sera in blocking buffer (PBS 1X + 0.05% Tween 20 + 3% Bovine

Serum Albumin) were incubated with purified therapeutic phage coated to immunoassay plates. For coated plate preparation, wells were incubated for 24 h either with coating buffer (carbonate-bicarbonate, pH 9.6) alone or containing purified phage diluted to 5×10^9 PFU/mL. For specific detection, Horseradish Peroxidase (HRP) conjugated anti-human IgM (Invitrogen A18841) and Ig-G (Invitrogen A18823) secondary antibodies (diluted 1:10,000) were used and colorimetric reaction was produced by the addition of 3,3',5,5'-tetramethylbenzidine substrate. After stopping the reaction with 2 N of H_2SO_4 , plates absorbance at 450 nm (signal) and 570 nm (background) were measured and the difference between both values was plotted against the serum dilution and fitted with a logistic curve.

Evaluation of phage-antibiotic interactions

When possible, phage-antibiotic interactions were evaluated as previously described,⁵⁰ with minor modifications. These assays were performed using the last clinical isolate before phage treatment initiation. Bacterial growth was evaluated in 96-well plates by optical density (OD) at 620 nm in the presence of the phage at different MOI, different concentrations of antibiotics and the combination of both. The MOI range evaluated was 10–0.001 and the antibiotic concentration was 0.008/0.4–0.00008/0.004 mg/mL for cotrimoxazole (trimethoprim/sulfamethoxazole), and 0.01–0.0001 mg/mL for linezolid. The antibiotic range was preliminarily assayed with 10-fold serial dilutions, to select those in which bacteria showed different susceptibility. Phage, antibiotic, or combinations of both, were added in triplicate to wells containing 10^5 CFU of bacteria to a final volume of 200 μL and growth kinetics were evaluated for 21 h at 37°C. Each plate contained one triplicate of bacterial control without phage or antibiotic, and one triplicate of LB media control, only phage control and only antibiotic control.

QUANTIFICATION AND STATISTICAL ANALYSIS

Mean and standard deviation were determined using GraphPad Prism (version 8.4.2, GraphPad Software, Boston, Massachusetts USA, www.graphpad.com). This software was also used for performing logistic curve fitting for ELISA data.